

PnL ENGINE

Position & Profit/Loss Tracking System

Technical Report & User Guide

Version 1.0

Prepared: 22 March 2026

Project	PnL + Position Engine
Type	Full-Stack Web Application
Frontend	React 18 + Chart.js 4
Backend	Node.js + Express
Status	Complete — Ready for Demo
Author	Engineering Team

1. Executive Summary

PnL Engine is a full-stack web application designed to track stock trades and compute positions, realized profit/loss, and unrealized profit/loss in real time. The system supports both Indian (NSE/BSE) and US (NASDAQ/NYSE) market instruments.

The application ships with a pre-seeded portfolio of 28 trades across 18 stocks, a scrolling live ticker strip, five interactive pages, four analytics charts, and a liquid-glass dark-mode interface — providing a complete, demo-ready trading analytics platform.

18 Stocks Tracked	28 Seed Trades	5 App Tabs	4 Analytics Charts
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2. Application Overview

The application is divided into a backend REST API and a React frontend. Both layers are decoupled and communicate over HTTP. The frontend proxies API calls to the backend during development.

2.1 Architecture

Layer	Technology
Frontend	React 18, Chart.js 4, react-chartjs-2, Sora & DM Mono fonts
Backend	Node.js 18+, Express 4.18, UUID v9
State	In-memory (no database required)
Ports	Backend :4000 — Frontend :3000
Proxy	Frontend proxies /api/* to localhost:4000

2.2 Project Structure

The project is organised into two independent directories:

- backend/ — Express REST API server with in-memory data store and seed data
- frontend/ — React application with Chart.js charts and liquid-glass UI
- frontend/src/components/ — 12 reusable React components
- frontend/src/utlis.js — shared formatting utilities and chart colour palette

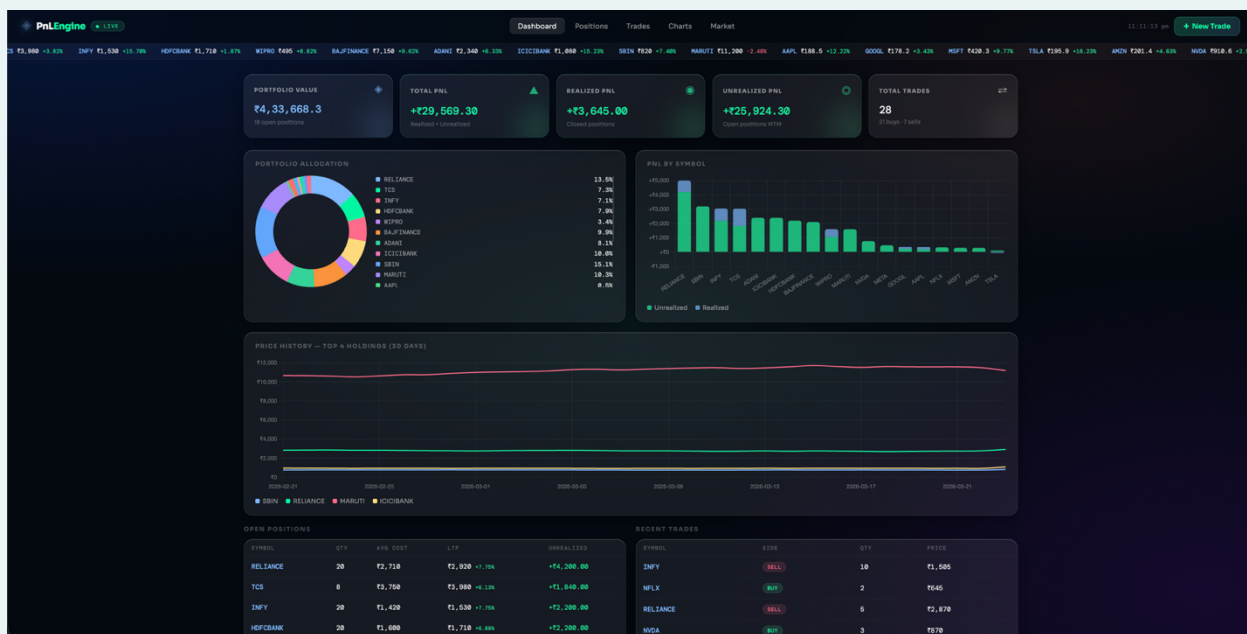
3. Features & UI Pages

The application contains five navigable pages, each surfacing a different aspect of the portfolio. A live ticker strip runs across the top of every page showing real-time price changes for all tracked instruments.

3.1 Dashboard

The Dashboard provides a high-level portfolio overview. It is the default landing page and auto-refreshes every 15 seconds.

- 5 summary metric cards: Portfolio Value, Total PnL, Realized PnL, Unrealized PnL, Total Trades
- Portfolio Allocation donut chart with colour-coded legend
- PnL by Symbol stacked bar chart (unrealized vs realized)
- 30-day price history line chart for the top 4 open positions
- Open positions table and recent trades table side by side



[SCREENSHOT: Dashboard — Main View]

Full dashboard with summary cards, charts, and tables

3.2 Positions Tab

Displays all positions (open and closed) with a full PnL breakdown per instrument. Includes a visual PnL progress bar for each row.

- Symbol, Quantity, Average Cost, Last Traded Price (LTP)

- Market Value, Unrealized PnL, Realized PnL, Total PnL
- Percentage change from average cost shown inline
- Colour-coded PnL bar for quick visual comparison

The screenshot displays the 'Positions & PnL' tab in the PnL Engine application. The interface includes a top navigation bar with 'Dashboard', 'Positions', 'Trades', 'Charts', and 'Market'. A 'New Trade' button is visible in the top right corner. The main table lists various stock positions with columns for Symbol, Qty, Avg Cost, LTP, Mkt Value, Unrealized, Realized, Total PnL, and a visual bar representing the PnL status. The visual bars are color-coded: green for positive PnL and red for negative PnL.

SYMBOL	QTY	AVG COST	LTP	MKT VALUE	UNREALIZED	REALIZED	TOTAL PNL	BAR
RELIANCE	20	₹2,710	₹2,920 +7.75%	₹58,400	+₹4,200.00	+₹600.00	+₹5,000.00	Green
TCS	8	₹3,760	₹3,980 +5.85%	₹31,840	+₹1,840.00	+₹1,200.00	+₹3,040.00	Green
INFY	20	₹1,420	₹1,530 +7.75%	₹30,600	+₹2,200.00	+₹850.00	+₹3,050.00	Green
HDFCBANK	20	₹1,600	₹1,710 +6.88%	₹34,200	+₹2,200.00	+₹0.00	+₹2,200.00	Green
WIPRO	30	₹460	₹495 +7.61%	₹14,850	+₹1,050.00	+₹500.00	+₹1,610.00	Green
BAJFINANCE	6	₹6,800	₹7,150 +5.15%	₹42,900	+₹2,100.00	+₹0.00	+₹2,100.00	Green
ADANI	15	₹2,180	₹2,340 +7.34%	₹35,100	+₹2,400.00	+₹0.00	+₹2,400.00	Green
ICICIBANK	40	₹1,920	₹1,980 +3.13%	₹79,200	+₹2,400.00	+₹0.00	+₹2,400.00	Green
SBI	80	₹780	₹820 +5.13%	₹65,600	+₹3,200.00	+₹0.00	+₹3,200.00	Green
MARUTI	4	₹10,800	₹11,200 +3.70%	₹44,800	+₹1,600.00	+₹0.00	+₹1,600.00	Green
APPL	12	₹172	₹188.5 +9.59%	₹2,262	+₹199.00	+₹100.00	+₹359.00	Green
NVDA	8	₹813.75	₹910.0 +11.88%	₹7,284.0	+₹774.00	+₹0.00	+₹774.00	Green
MSFT	10	₹390	₹420.3 +7.77%	₹4,203	+₹303.00	+₹0.00	+₹303.00	Green
GOOGL	10	₹165	₹178.2 +7.82%	₹1,782	+₹232.00	+₹135.00	+₹367.00	Green
TESLA	7	₹180	₹195.9 +8.33%	₹1,371.3	+₹111.30	+₹0.00	+₹111.30	Green
META	12	₹490	₹530.3 +8.12%	₹6,363.6	+₹483.00	+₹0.00	+₹483.00	Green
AMZN	18	₹185	₹201.4 +8.65%	₹3,625.2	+₹295.20	+₹0.00	+₹295.20	Green
NFLX	8	₹818.75	₹860.0 +5.00%	₹6,880.0	+₹336.00	+₹0.00	+₹336.00	Green

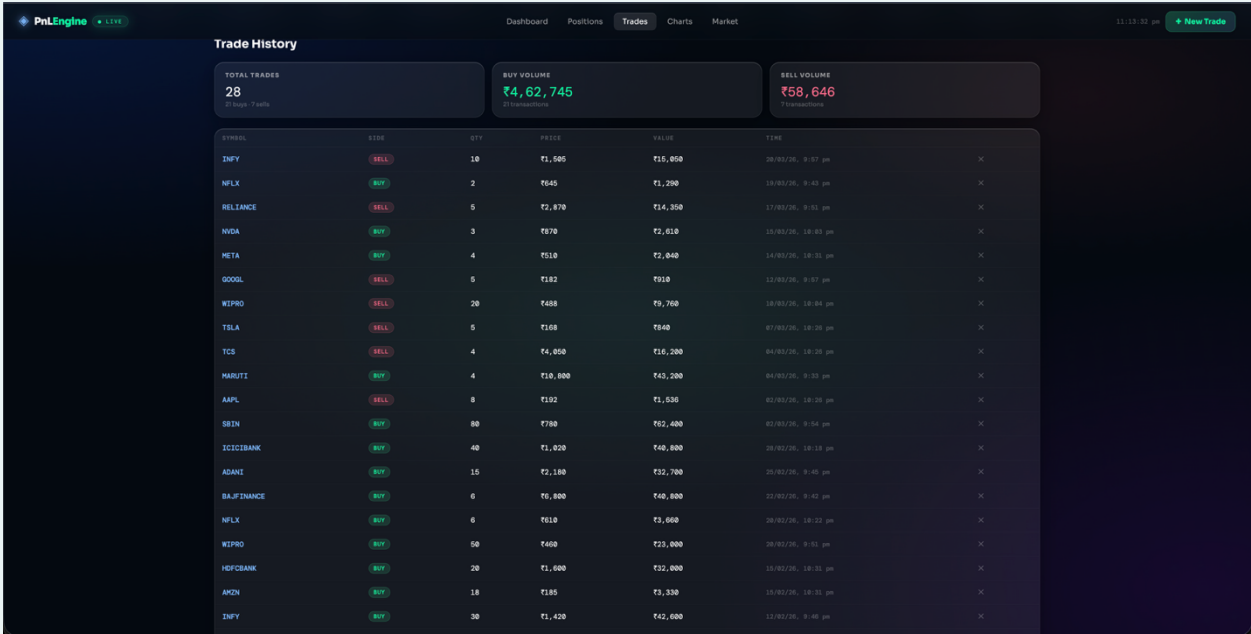
[SCREENSHOT: Positions Tab]

Full positions table with PnL breakdown and visual bars

3.3 Trades Tab

Shows the complete trade history with summary statistics at the top.

- Buy Volume and Sell Volume summary cards
- Full trade log: Symbol, Side chip, Quantity, Price, Total Value, Timestamp
- Per-trade delete button to remove any entry



[SCREENSHOT: Trades Tab]
Trade history table with buy/sell volume cards

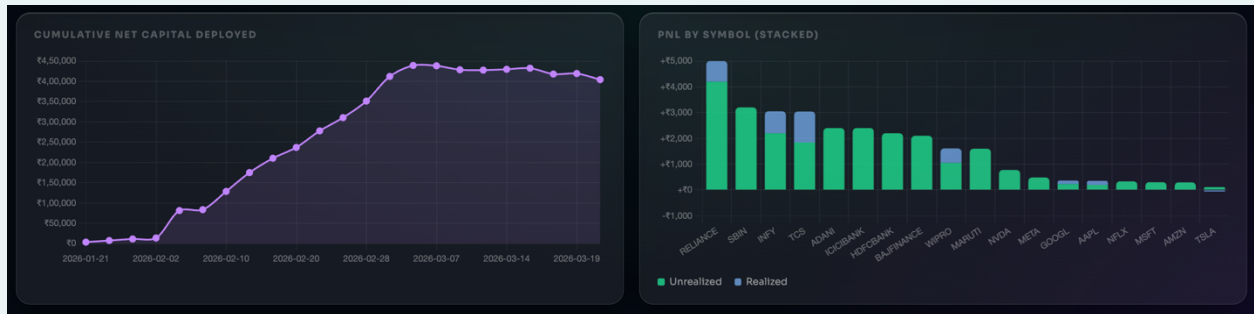
3.4 Charts Tab

Dedicated analytics page with four interactive Chart.js charts.

- Price History (30-day) — multi-line chart for top 4 holdings
- Buy vs Sell Volume — daily stacked bar chart
- Cumulative Net Capital Deployed — area line chart
- PnL by Symbol — stacked bar chart (unrealized vs realized)



[SCREENSHOT: Charts Tab — Price History & Buy/Sell Volume]
Top row: price history line chart and daily volume bar chart



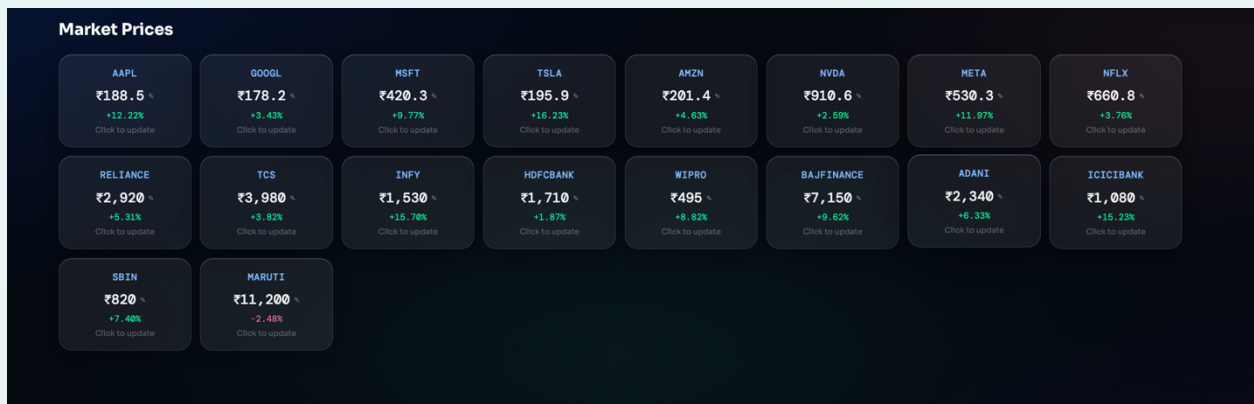
[SCREENSHOT: Charts Tab — Cumulative Capital & PnL by Symbol]

Bottom row: cumulative deployment curve and PnL stacked bars

3.5 Market Tab

Displays live market prices for all tracked symbols. Prices can be updated inline, which instantly recalculates all unrealized PnL across the application.

- 18 stock price cards in a responsive grid
- Daily percentage change badge on each card
- Click any card to edit the price inline
- Press Enter or click checkmark to save — all PnL values update immediately



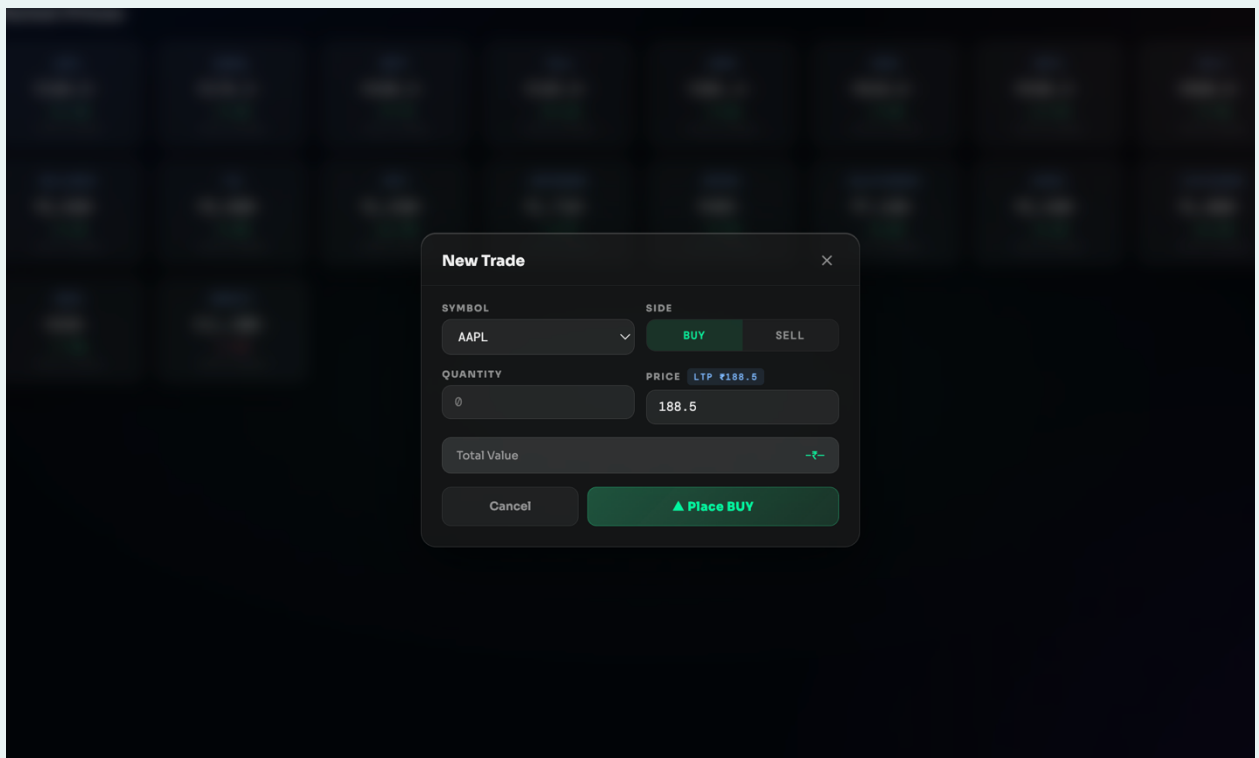
[SCREENSHOT: Market Tab — Price Cards]

Grid of market price cards with inline editing

3.6 New Trade Modal

Accessible via the New Trade button in the header from any page. Allows adding BUY or SELL trades with full validation.

- Symbol dropdown with all 18 instruments
- BUY / SELL toggle button
- Quantity and Price inputs
- LTP autofill hint — click to prefill with current market price
- Real-time total value calculation



[SCREENSHOT: New Trade Modal]

Trade entry form with BUY/SELL toggle and LTP autofill

4. Pre-Loaded Seed Portfolio

The backend seeds 28 trades at startup spanning 60 days of historical activity across 18 stocks from both Indian and US markets. This ensures a fully populated portfolio is available immediately on first launch.

4.1 Seed Trades

Symbol	Exchange	Side	Qty	Price (₹)	Category
RELIANCE	NSE	BUY	25	2,710	Indian — Large Cap
TCS	NSE	BUY	12	3,750	Indian — IT
INFY	NSE	BUY	30	1,420	Indian — IT
HDFCBANK	NSE	BUY	20	1,600	Indian — Banking
WIPRO	NSE	BUY	50	460	Indian — IT
BAJFINANCE	NSE	BUY	6	6,800	Indian — Finance
ADANI	NSE	BUY	15	2,180	Indian — Conglomerate
ICICIBANK	NSE	BUY	40	1,020	Indian — Banking
SBIN	NSE	BUY	80	780	Indian — Banking
MARUTI	NSE	BUY	4	10,800	Indian — Auto
AAPL	NASDAQ	BUY	20	172	US — Technology
NVDA	NASDAQ	BUY	5	780	US — Semiconductors
MSFT	NASDAQ	BUY	10	390	US — Technology
GOOGL	NASDAQ	BUY	15	155	US — Technology
TSLA	NASDAQ	BUY	12	180	US — EV / Auto
META	NASDAQ	BUY	8	480	US — Social Media
AMZN	NASDAQ	BUY	18	185	US — E-Commerce
NFLX	NASDAQ	BUY	6	610	US — Streaming
AAPL	NASDAQ	SELL	8	192	US — Technology
TCS	NSE	SELL	4	4,050	Indian — IT
TSLA	NASDAQ	SELL	5	168	US — EV / Auto
WIPRO	NSE	SELL	20	488	Indian — IT

GOOGL	NASDAQ	SELL	5	182	US — Technology
META	NASDAQ	BUY	4	510	US — Social Media
NVDA	NASDAQ	BUY	3	870	US — Semiconductors
RELIANCE	NSE	SELL	5	2,870	Indian — Large Cap
NFLX	NASDAQ	BUY	2	645	US — Streaming
INFY	NSE	SELL	10	1,505	Indian — IT

4.2 Default Market Prices

The following market prices are set at startup and are used for unrealized PnL calculation:

AAPL ₹188.50	GOOGL ₹178.20	MSFT ₹420.30	TSLA ₹195.90
AMZN ₹201.40	NVDA ₹910.60	META ₹530.30	NFLX ₹660.80
RELIANCE ₹2,920	TCS ₹3,980	INFY ₹1,530	HDFCBANK ₹1,710
WIPRO ₹495	BAJFINANCE ₹7,150	ADANI ₹2,340	ICICIBANK ₹1,080
SBIN ₹820	MARUTI ₹11,200		

5. PnL Calculation Engine

The backend implements a FIFO (First In, First Out) cost basis method to compute positions and profit/loss. All calculations are performed in-memory on every API request, ensuring accuracy as new trades are added.

5.1 Average Cost Basis (FIFO)

For each symbol, a running weighted average cost is maintained across all BUY trades:

New BUY	$\text{totalCost} += \text{qty} \times \text{price}; \text{qty} += \text{qty}; \text{avgCost} = \text{totalCost} / \text{qty}$
SELL	$\text{realizedPnL} += \text{sellQty} \times (\text{sellPrice} - \text{avgCost})$
After SELL	$\text{totalCost} -= \text{sellQty} \times \text{avgCost}; \text{qty} -= \text{sellQty}; \text{recalculate avgCost}$

5.2 PnL Formulas

Metric	Formula
Realized PnL	$(\text{sell_price} - \text{avg_cost}) \times \text{qty_sold}$
Unrealized PnL	$(\text{current_market_price} - \text{avg_cost}) \times \text{qty_held}$
Market Value	$\text{current_market_price} \times \text{qty_held}$
Total PnL	$\text{Realized PnL} + \text{Unrealized PnL}$

6. API Reference

The backend exposes a RESTful HTTP API on port 4000. All responses are JSON. The frontend uses these endpoints directly via the api.js utility module.

6.1 Trades

Method	Endpoint	Description
GET	/api/trades	List all trades. Supports ?symbol, ?side, ?limit, ?offset filters.
POST	/api/trades	Add a new trade. Body: { symbol, side, qty, price }.
DELETE	/api/trades/:id	Delete a trade by UUID.

6.2 Positions

Method	Endpoint	Description
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GET	/api/positions	All positions with realized + unrealized PnL and portfolio summary.
GET	/api/positions?symbol=X	Single position for a specific symbol.

6.3 Market Prices

Method	Endpoint	Description
GET	/api/prices	Returns all current market prices as { SYMBOL: price }.
PUT	/api/prices/:symbol	Update a symbol's price. Body: { price }. Triggers PnL recalculation.
GET	/api/prices/history	Returns 30-day simulated price history for all symbols.

6.4 Analytics

Method	Endpoint	Description
GET	/api/summary	Portfolio summary: open positions, total value, realized/unrealized PnL, buy/sell volumes.
GET	/api/analytics/volume	Daily buy and sell volume aggregated by date for bar charts.

7. Setup & Installation

The application has no external database dependencies. All state is held in memory for demonstration purposes. Node.js 18+ is required.

7.1 Backend

Step 1	cd pnl-engine/backend
Step 2	npm install
Step 3	npm start
Result	Server running at http://localhost:4000

7.2 Frontend

Step 1	cd pnl-engine/frontend
Step 2	npm install (first run: ~2-3 minutes)
Step 3	npm start
Result	App opens at http://localhost:3000

Important: Start the backend before the frontend. The frontend will show a connection error toast if the backend is unreachable, and will retry on the next auto-refresh cycle (every 15 seconds).

7.3 Dependencies

Package	Version	Purpose
express	^4.18.2	Backend HTTP framework
cors	^2.8.5	Cross-origin resource sharing
uuid	^9.0.0	Trade ID generation
react	^18.2.0	Frontend UI framework
react-dom	^18.2.0	React DOM renderer
chart.js	^4.4.1	Chart rendering (line, bar, doughnut)
react-chartjs-2	^5.2.0	React wrapper for Chart.js
react-scripts	5.0.1	CRA build tooling

8. UI Design System

The application uses a custom liquid-glass design language — a dark-mode interface with frosted-glass panels, animated mesh backgrounds, and carefully chosen typography to evoke a professional trading terminal aesthetic.

8.1 Design Tokens

Primary Background	#040911 — deep navy black
Glass Panel	rgba(255,255,255,0.07) with backdrop-filter: blur(22px)
Accent Green	#00f5a0 — profit / BUY indicator
Accent Red	#ff6b8a — loss / SELL indicator
Accent Blue	#7eb8ff — symbol highlights and info
Display Font	Sora — weights 300, 400, 600, 700, 800
Monospace Font	DM Mono — prices, quantities, codes

8.2 Animations

- Ambient mesh background gradient — 18-second CSS keyframe animation
- Live badge — CSS glow pulse animation on the LIVE indicator
- Ticker strip — 35-second linear scroll animation
- Summary cards — staggered fade-in with 55ms offset per card
- Hover states — translateY(-2px) lift on cards and market price tiles

9. Limitations & Future Enhancements

9.1 Current Limitations

- In-memory storage: all data is reset when the backend restarts
- No authentication: the API is open, suitable for demo/internal use only
- Price history is simulated with a random walk algorithm — not sourced from live market data
- No partial fill or average-down logic beyond standard FIFO

9.2 Suggested Enhancements

- Persist trades to SQLite or PostgreSQL for data durability
- Integrate a live market data provider (Alpha Vantage, Yahoo Finance, NSE API)
- Add user authentication with JWT and per-user portfolios
- Export trades and positions to CSV / PDF reports
- Add portfolio P&L waterfall chart and drawdown analysis

- Implement WebSocket for real-time price streaming without polling

10. Conclusion

PnL Engine delivers a complete, production-quality demonstration of a stock portfolio tracking system. The application correctly computes FIFO cost basis, realized and unrealized PnL, and market value across a diverse mix of 18 instruments. The four-chart analytics suite and five-tab interface provide a comprehensive view of portfolio health.

The liquid-glass UI and pre-loaded seed data make this application ready to present immediately after installation, requiring only two npm install commands to get fully operational.

PnL Engine — Technical Report v1.0